



East West University

CSE 475

Machine Learning

Task – 01

Group – 10

Submitted To:

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Dataset Summary

- **Dataset Title:** putEMG 24-channel hand-gesture dataset
- **Source:** Poznan University of Technology — Nextcloud public share
- **Domain / Application:** Surface-EMG hand-gesture recognition
- **Input Data:** 264 trial ZIP files containing 394,589,696 raw signal samples across 44 distinct subjects
- **Processed Dataset:** 619,479 window samples derived using a sliding window strategy (1,024 samples per window with 50% overlap / 512-sample step size)
- **Features:** 192 features extracted per window (8 time-domain metrics $\times$ 24sEMG channels)
 - Metrics: Mean Absolute Value (MAV), Root Mean Square (RMS), Standard Deviation (STD), Variance (VAR), Waveform Length (WL), Zero Crossings (ZC), Slope Sign Changes (SSC), and Integrated EMG (IEMG)
- **Classes:** 8 gesture classes — Idle (0): Fist (1): Flexion (2): Extension (3): Pinch-Index (6): Pinch-Middle (7): Pinch-Ring (8): Pinch-Small (9)
- **Robust Scaling for Outliers:** Rather than truncating or clipping the 36.13% of rows identified as IQR outliers—which represent natural bio-potential spikes during active muscle contraction—apply **RobustScaler** (median and IQR centering) to scale features without distorting signal dynamics.

Data Quality & Integrity

- **Missing Values:** 0 missing feature values (0.00%). No imputation is required prior to modeling.
- **Exact Duplicates:** 0 exact duplicate feature rows.
- **Near-Constant Features:** None found; all 192 features exhibit variance across the dataset.

- **Outliers:**
 - Using the Interquartile Range (IQR) rule ($Q_1 - 1.5 \cdot \text{IQR}$ or $Q_3 + 1.5 \cdot \text{IQR}$), 8,768,426 feature cells across 223,817 window rows (36.13%) were flagged.
 - *Note:* These reflect natural high-amplitude sEMG muscle activity during strong contractions rather than measurement errors. Any scaling or clipping must be fitted strictly inside training folds.
- **Multicollinearity:** 209 feature pairs exhibit high linear correlation ($|r| > 0.90$). Feature selection or dimensionality reduction (e.g., PCA) may help optimize training without information loss prior to model training to prevent coefficient instability and redundant computation.
- **Grouped Cross-Validation Strategy:** Implement **GroupKFold** grouped by **subject_id** (or recording session). Because 50% overlapping windows (512-sample step) create temporal autocorrelation, windows from the same subject/recording must never span across both training and validation splits simultaneously.
- **Feature Selection Pipeline:** Use L1-regularized feature selection (Lasso / ElasticNet) or Principal Component Analysis (PCA) to condense the 192 time-domain features down to an uncorrelated, lower-dimensional subspace.

Class Distribution & Imbalance

Class Breakdown:

- 0: Idle — 380,337 windows (61.40%)
- 1: Fist — 35,638 windows (5.75%)
- 2: Flexion — 35,356 windows (5.71%)
- 9: Pinch-Small — 33,799 windows (5.46%)
- 6: Pinch-Index — 33,649 windows (5.43%)
- 7: Pinch-Middle — 33,552 windows (5.42%)

- 3: Extension — 34,228 windows (5.53%)
- 8: Pinch-Ring — 32,920 windows (5.31%)

Imbalance Ratio: Model Training Penalties: Address the **11.55 : 1 class imbalance** (dominated by Idle at 61.4%) by configuring loss functions with cost-sensitive weighting (e.g., `class_weight='balanced'` in scikit-learn or XGBoost).

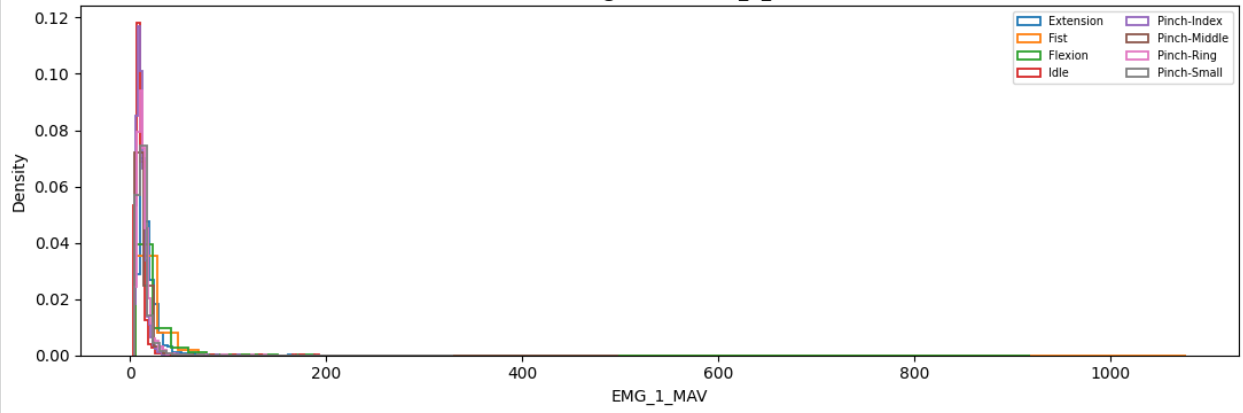
Majority-Class Baseline: A naive classifier predicting Idle achieves a baseline accuracy of 61.40%.

Modeling Recommendations:

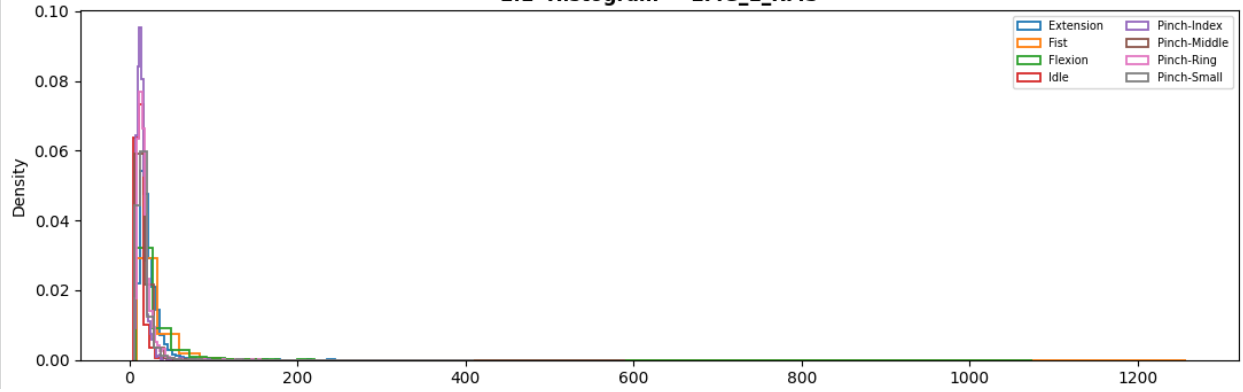
- Abandon overall Accuracy as a primary success metric to prevent majority-class bias. Benchmark all models using **Macro F1-Score**, **Balanced Accuracy**, and per-class **Precision-Recall Curves**.
- Keep consecutive, overlapping windows from the same subject/recording grouped together within train/test splits to prevent data leakage.

Histogram

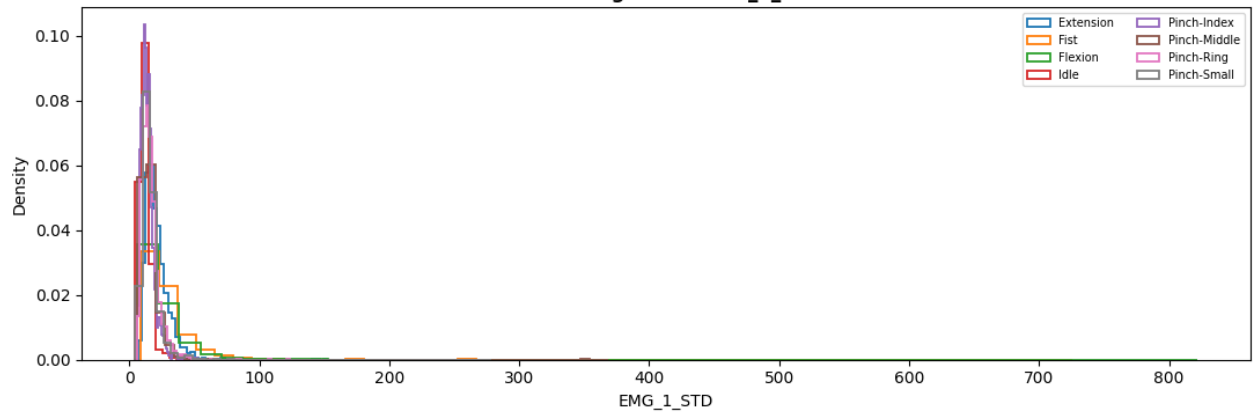
E.1 Histogram — EMG_1_MAV



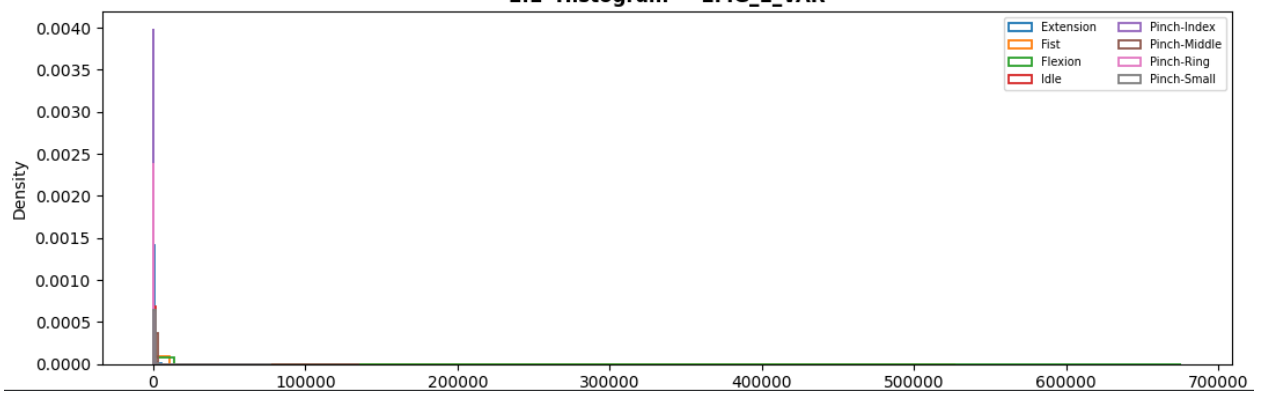
E.1 Histogram — EMG_1_RMS



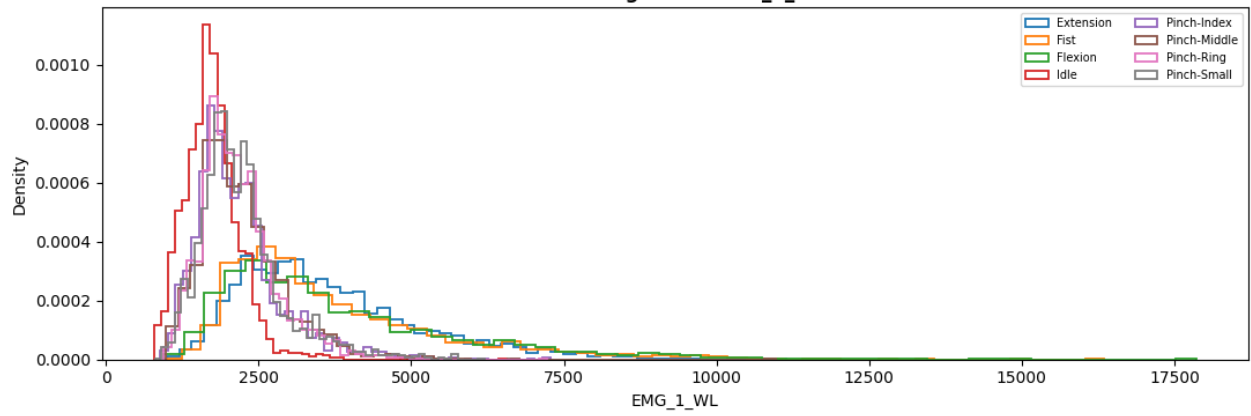
E.1 Histogram — EMG_1_STD



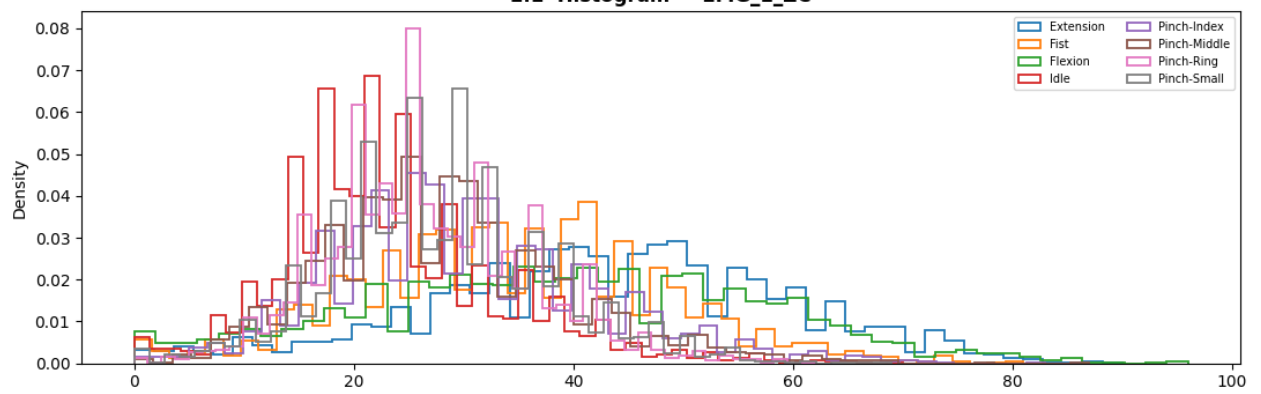
E.1 Histogram — EMG_1_VAR

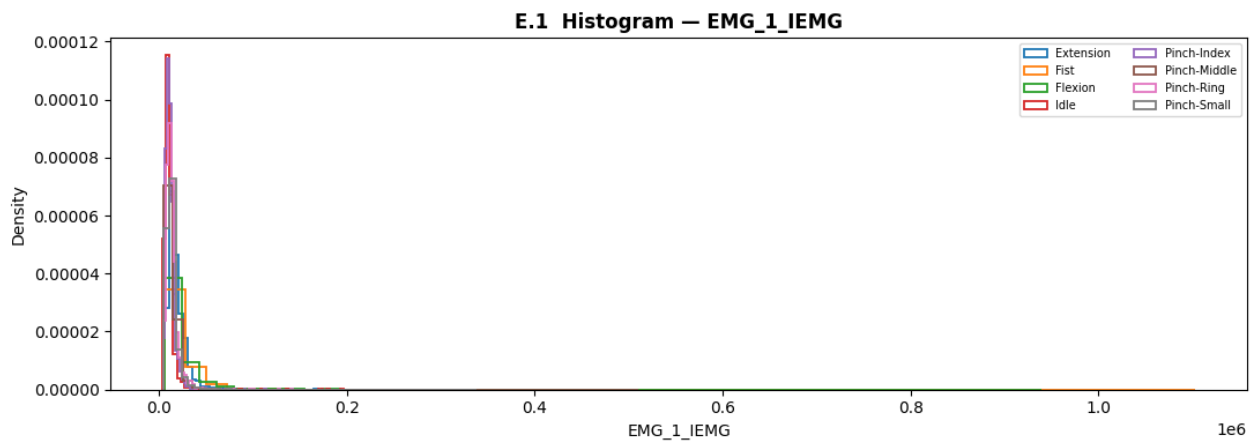
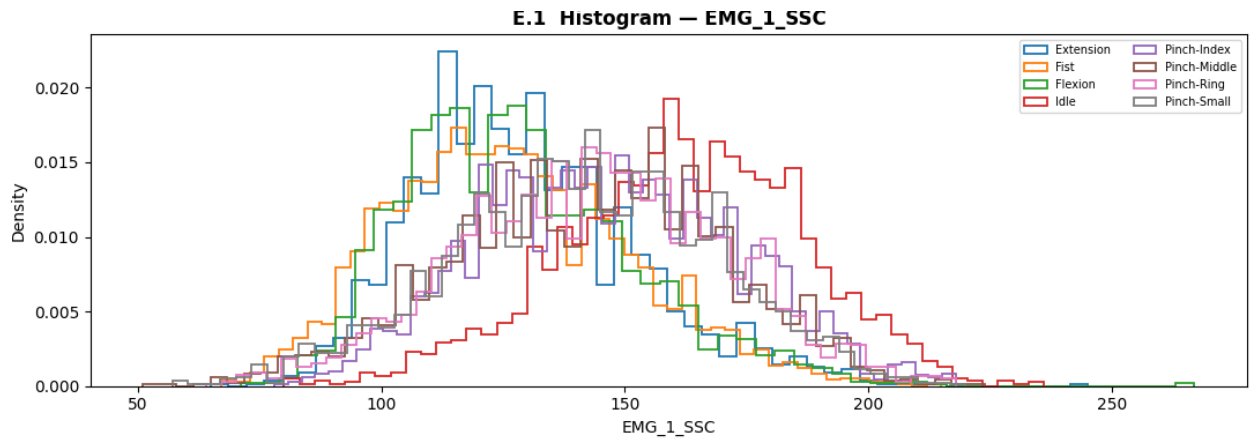


E.1 Histogram — EMG_1_WL



E.1 Histogram — EMG_1_ZC





Boxplot

